

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- No additional service items required.

Setup

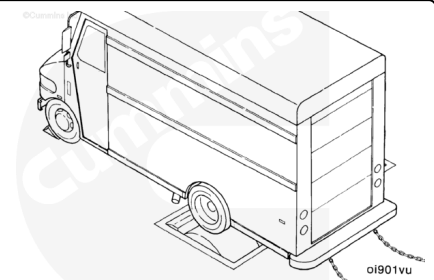
The performance of an engine installed in on-highway vehicles can be tested on a chassis dynamometer.

Note : Because of live driveline efficiency and engine-driven accessories, the engine horsepower when measured at the rear wheels will be reduced by approximately:

- 20 percent for single-axle vehicles
- 25 percent for tandem-axle vehicles
- 30 percent for tri-axles vehicles
- 35 percent for recreational vehicles.

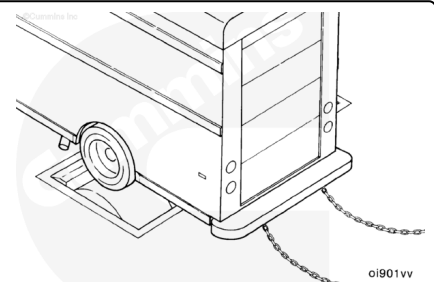
These specifications apply to drive axles **only**, **not** tag axles.

Note : These percentages are used for engine run-in **only** and are **not** to be used as absolute figures.



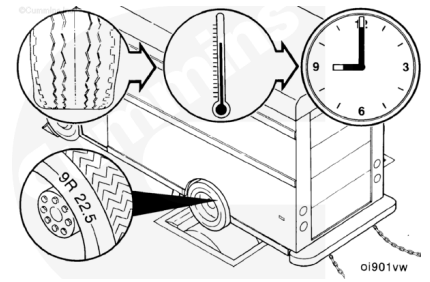
⚠ WARNING ⚠

Follow all the OEM's safety precautions before installing or operating a vehicle on a chassis dynamometer. Failure to do so can cause damage to the vehicle and/or cause personal injury.



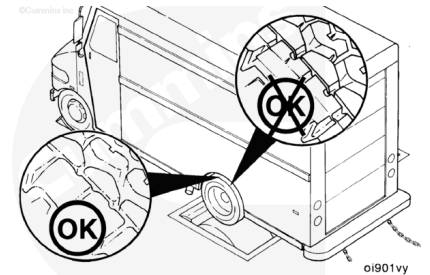
⚠ CAUTION ⚠

Low-profile tires are more sensitive to heat than bias ply tires. Excessive operating time at full load can damage the tires due to overheating. Check the tire manufacturer's recommendations for the maximum allowable chassis dynamometer operating time.

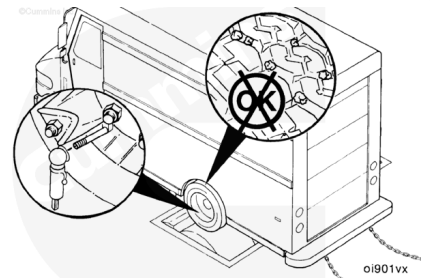


While operating the chassis dynamometer, follow the general safety precautions listed below:

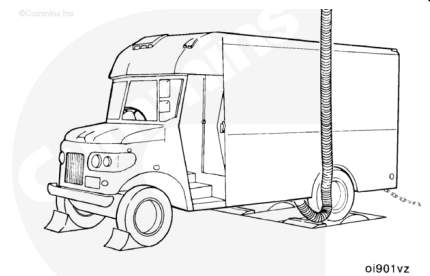
- Use tires that have more than 160 km [100 mi] of use. Do **not** use new tires.
- Do **not** use recapped tires or tires of different sizes or designs.



- Make sure the tires are inflated to manufacturer specifications.
- Remove all rocks or other materials from the treads of the tires rotating on the dynamometer rollers.



- Make sure there is correct overhead clearance for exhaust stacks, air deflectors, or other attachments above the cab.



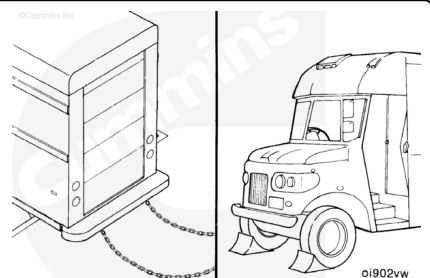
⚠ CAUTION ⚠

To prevent damage to the chassis dynamometer, make sure there is enough slack in the tie-down chains.

Carefully position the vehicle on the rollers.

Attach the tie-down chains to the rear of the vehicle.

Place wheel chocks in front of the front tires.

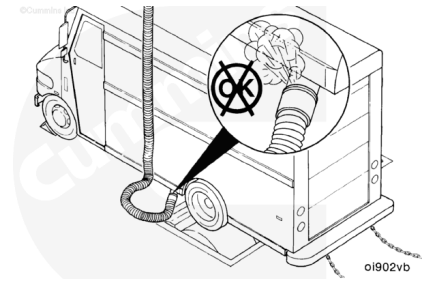


⚠ WARNING ⚠

Some exhaust gas constituents may be toxic and/or carcinogenic. Make sure the ventilation hose does not leak.

Adjust the vehicle and dynamometer room exhaust system to make sure all the exhaust gases are removed from the room.

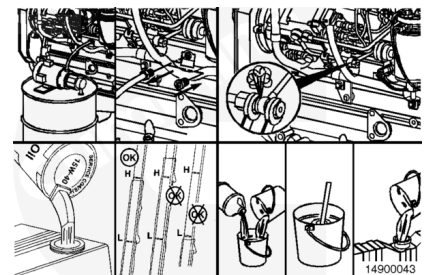
Read the chassis dynamometer and vehicle manufacturer's recommendations and specifications for testing procedures.



Test

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The following procedure assumes that the lubricating oil and fuel systems were correctly primed, the dipstick calibrated, and the engine filled to the correct levels with lubricating oil and coolant during installation of the engine into the chassis. If these systems were **not** serviced during installation of the engine.

Use the following procedures for instructions on priming the lubricating oil and the fuel system and calibrating the dipstick.

- Use the following procedure in the Service Manual, ISF2.8 CM2220 F101. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/197/197-014-006-tr.html>)
- Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/119/119-014-006.html>)
- Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/119/119-014-006.html>)

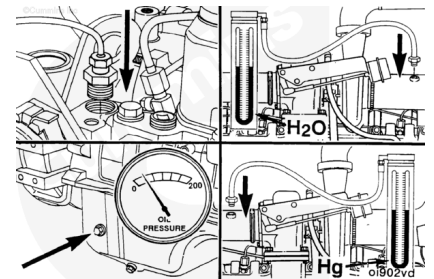
- Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/118/118-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/53/53-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/40/40-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/45/45-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/41/41-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/44/44-014-006.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/40/40-014-006.html>)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/261/261-014-006.html>)
- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/409/409-014-006.html>)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/459/459-014-006.html>)

- Use the following procedure in the F2.8 CM2620 F135B Service Manual, Bulletin 5504616. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/593/593-014-006.html)
- Use the following procedure in the F2.8 CM2620 F136B Service Manual, Bulletin 5504621. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/595/595-014-006.html)
- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/1365/1365-014-006.html)

See the Operation and Maintenance Manual for instructions on filling the lubricating oil and the cooling systems.

The number of instruments and gauges required to perform a chassis dynamometer test will vary according to type and the capability of the test equipment used.

See the service tools listed at the beginning of this section.



To correctly monitor engine performance, record the following parameters:

- Lubricating oil pressure (vehicle instrument panel)
- Engine speed (rpm) (vehicle instrument panel)
- Wheel horsepower (whp) (dynamometer controls)
- Blowby: Refer to Procedure 014-010 in Section 14.
(/qs3/pubsys2/xml/en/procedures/100/100-014-010.html)
- Exhaust back pressure:
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 011-009 in Section 14.
(/qs3/pubsys2/xml/en/procedures/197/197-011-009-tr.html)
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/119/119-011-009.html)

- Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/118/118-011-009.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/53/53-011-009.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/40/40-011-009-tr.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/45/45-011-009.html)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/41/41-011-009.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/44/44-011-009.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 011-009 in Section 11.
(/qs3/pubsys2/xml/en/procedures/40/40-011-009.html)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/302/302-011-009.html)

- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/421/421-011-009.html)

- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/459/459-011-009.html)

- Use the following procedure in the F2.8 CM2620 F135B Service Manual, Bulletin 5504616. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/593/593-011-009.html)

- Use the following procedure in the F2.8 CM2620 F136B Service Manual, Bulletin 5504621. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/595/595-011-009.html)

- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/1365/1365-011-009.html)

- Coolant temperature (vehicle instrument panel):

- Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 008-018 in Section 18.

(/qs3/pubsys2/xml/en/procedures/197/197-008-018-tr.html)

- Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN, Bulletin 4022178. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/119/119-008-018-tr.html)

- Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/118/118-008-018.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/53/53-008-018-tr.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/53/53-008-018-tr.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/44/44-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-018.html)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/261/261-008-018-tr.html)

- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/409/409-008-018-tr.html)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/459/459-008-018-tr.html)
- Use the following procedure in the F2.8 CM2620 F135B Service Manual, Bulletin 5504616. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/593/593-008-018.html)
- Use the following procedure in the F2.8 CM2620 F136B Service Manual, Bulletin 5504621. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/595/595-008-018.html)
- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1365/1365-008-018-tr.html)
- Coolant pressure:
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/197/197-008-018-tr.html)
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/119/119-008-018-tr.html)
 - Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/53/53-008-018-tr.html)
 - Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 008-

018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/44/44-008-018.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-018.html)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/261/261-008-018-tr.html)
- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/409/409-008-018-tr.html)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/459/459-008-018-tr.html)
- Use the following procedure in the F2.8 CM2620 F135B Service Manual, Bulletin 5504616. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/593/593-008-018.html)

- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/1365/1365-008-018-tr.html)

- Intake manifold pressure:

- Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 019-159 in Section 19.

(/qs3/pubsys2/xml/en/procedures/197/197-019-159-tr.html)

- Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 010-031 in Section 10.

(/qs3/pubsys2/xml/en/procedures/119/119-010-031.html)

- Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 010-057 in Section 10.

(/qs3/pubsys2/xml/en/procedures/118/118-010-057.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 010-057 in Section 10.

(/qs3/pubsys2/xml/en/procedures/53/53-010-057.html)

- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 010-057 in Section 10.

(/qs3/pubsys2/xml/en/procedures/40/40-010-057.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 010-057 in Section 10.

(/qs3/pubsys2/xml/en/procedures/45/45-010-057.html)

- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 010-057 in Section 10.

(/qs3/pubsys2/xml/en/procedures/41/41-010-057.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 010-057 in Section 10.
(/qs3/pubsys2/xml/en/procedures/44/44-010-057.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 010-057 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-057.html)
- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 019-159 in Section 19.
(/qs3/pubsys2/xml/en/procedures/409/409-019-159.html)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 019-159 in Section 19.
(/qs3/pubsys2/xml/en/procedures/459/459-019-159.html)
- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 019-159 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1365/1365-019-159.html)
- Inlet air restriction:
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/197/197-010-031-tr.html)
 - Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/119/119-010-031.html)
 - Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/118/118-010-031.html)
 - Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin

3666193. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/53/53-010-031.html)

- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/100/100-010-031.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/45/45-010-031.html)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/100/100-010-031.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/44/44-010-031.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-031.html)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/261/261-010-031.html)
- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 010-031 in Section 10.
(/qs3/pubsys2/xml/en/procedures/409/409-010-031.html)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to

Procedure 010-031 in Section 10.

(/qs3/pubsys2/xml/en/procedures/459/459-010-031.html)

- Use the following procedure in the F2.8 CM2620 F135B Service Manual, Bulletin 5504616. Refer to Procedure 010-031 in Section 10.

(/qs3/pubsys2/xml/en/procedures/593/593-010-031.html)

- Use the following procedure in the F2.8 CM2620 F136B Service Manual, Bulletin 5504621. Refer to Procedure 010-031 in Section 10.

(/qs3/pubsys2/xml/en/procedures/595/595-010-031.html)

- Use the following procedure in the R2.8 CM2220 R102C Service Manual, Bulletin 6457240. Refer to Procedure 010-031 in Section 10.

(/qs3/pubsys2/xml/en/procedures/1365/1365-010-031.html)

- Fuel supply pressure:

- Use the following procedure in the Service Manual, ISF2.8 CM2220 F101, Bulletin 4310846. Refer to Procedure 006-051 in Section 6.

(/qs3/pubsys2/xml/en/procedures/197/197-006-051-tr.html)

- Use the following procedure in the Service Manual, ISF2.8 CM2220 E and ISF2.8 CM2220 AN Bulletin 4022178. Refer to Procedure 005-016 in Section 5.

(/qs3/pubsys2/xml/en/procedures/119/119-005-016-tr.html)

- Use the following procedure in the Service Manual, ISF3.8 CM2220 Bulletin 4021704. Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/118/118-006-024-tr.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISB and QSB5.9 Engines, Bulletin 3666193. Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/53/53-006-024.html)

- Use the following procedure in the Troubleshooting and Repair Manual, B3.9, B4.5, and B5.9 Series Engines, Bulletin 3666087. Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/40/40-006-024.html)

- Use the following procedure in the Troubleshooting and Repair Manual, ISC, ISCe, QSC8.3, ISL and QSL9 Engines, Bulletin 4021418. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/45/45-006-024.html)
- Use the following procedure in the Troubleshooting and Repair Manual, C Series, Bulletin 3666003. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/41/41-006-024.html)
- Use the following procedure in the Troubleshooting and Repair Manual, ISB, ISBe 4 and 6 Cylinder, Bulletin 4021271. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/44/44-006-024.html)
- Use the following procedure in the Troubleshooting and Repair Manual, B Series, Bulletin 3810207. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/40/40-006-024.html)
- Use the following procedure in the QSL9 CM2350 L102 and L9 CM2450 L181C Service Manual, Bulletin 4332796. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/261/261-006-024-tr.html)
- Use the following procedure in the QSL9 CM2350 L118 and L9 CM2450 L182C Service Manual, Bulletin 4388626. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/409/409-006-024.html)
- Use the following procedure in the L9 CM2350 L116B Service Manual, Bulletin 5411330. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/459/459-006-024.html)

